T-101 to C-201 feed

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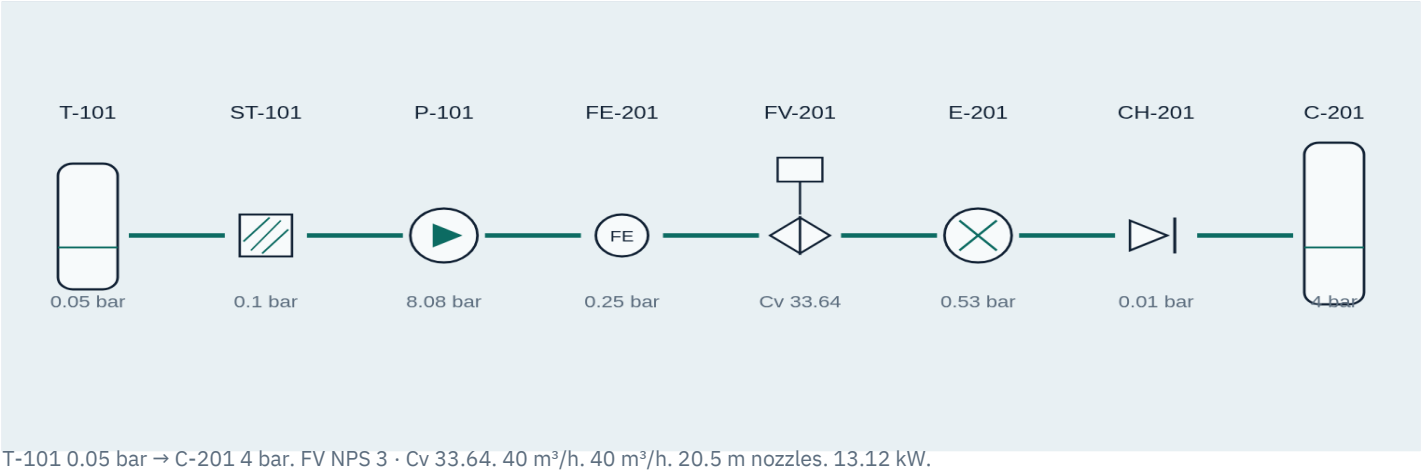
Column feed · Methanol

Pump differential	Control valve Cv	Valve drop	Pump electric
8.08 bar	**33.64**	**1.5 bar**	**13.12 kW**



T-101 0.05 bar → C-201 4 bar. FV NPS 3 · Cv 33.64. 40 m³/h. 40 m³/h. 20.5 m nozzles. 13.12 kW.

1 Process properties

Fluid	Methanol
Temperature	25 °C
Flow	40 m³/h
Mass flow	31,680 kg/h
Density	792 kg/m³
Viscosity	0.59 cP
Vapour pressure	12.9 kPa abs
Critical pressure	8,090 kPa abs
Specific gravity	0.794
Atmosphere	1.013 bar abs

2 Boundaries

Source	T-101
Source pressure	0.05 bar g
Source elevation	1.5 m
Destination	C-201
Destination pressure	4 bar g
Destination elevation	22 m
Sum of item rises	20.5 m
Pipe roughness	0.046 mm
Static drop, fouled design	1.59 bar
Dynamic drop, fouled design	1.04 bar

3 Pump data

Tag	P-101
Basis	Sized on fouled design flow
Head curve	Centrifugal screen, shutoff about 12% above design
Differential	8.08 bar
Head	104 m
Hydraulic power	8.98 kW
Electric power	13.12 kW

Pump efficiency	72%
Motor efficiency	95%
NPSHa, fouled suction	12.21 m
NPSHr	2.5 m
NPSH margin	9.71 m
Annual energy	104,973.1 kWh · 30,442.2 AED

4 Control valve

Tag	FV-201
Style	Globe, equal percentage
Inlet pressure	6.2 bar
Outlet pressure	4.7 bar
Differential	1.5 bar
Kv	29.09
Cv	33.64
Screened body	NPS 3
Full-open Cv, screen	110
Design travel	69.71%
Authority	0.59
Liquid pressure recovery Fl	0.9
Choked	No
Specific gravity	0.794

5 Equipment and line items

Tag	Item	ΔP	P out
6"-P-1401	Pipe	-0.11 bar	0.16 bar
NPS 6 Sch 40 / STD. 8 m · elev -1.5 m · 0.6 m/s · 2 × 90° elbow LR, 1 × Gate valve, 1 × Sharp entrance			
ST-101	Strainer	0.1 bar	0.06 bar
NPS 6 Sch 40 / STD. Clean 0.05 bar · fouling ×2			
P-101	Pump	-8.08 bar	8.14 bar
Duty machine. Sized to hold destination pressure			
4"-P-1402	Pipe	0.02 bar	8.12 bar
NPS 4 Sch 40 / STD. 6 m · elev 0 m · 1.35 m/s · 2 × 90° elbow LR, 1 × Gate valve			
FE-201	Flow element	0.25 bar	7.87 bar
NPS 4 Sch 40 / STD. Clean 0.25 bar · fouling ×1			
4"-P-1403	Pipe	1.67 bar	6.2 bar
NPS 4 Sch 40 / STD. 45 m · elev +20.5 m · 1.35 m/s · 6 × 90° elbow LR, 2 × Gate valve, 1 × Ball valve			
FV-201	Control valve	1.5 bar	4.7 bar
globe NPS 3. Design drop 1.5 bar · Fl 0.9			
4"-P-1404	Pipe	0.14 bar	4.56 bar
NPS 4 Sch 40 / STD. 12 m · elev +1.5 m · 1.35 m/s · 3 × 90° elbow LR			
E-201	Exchanger	0.53 bar	4.03 bar
NPS 4 Sch 40 / STD. Clean 0.35 bar · fouling ×1.5			
CH-201	Check valve	0.01 bar	4.02 bar
NPS 4 Sch 40 / STD. gate · qty 1			
4"-P-1405	Pipe	0.02 bar	4 bar
NPS 4 Sch 40 / STD. 6 m · elev 0 m · 1.35 m/s · 1 × 90° elbow LR, 1 × Exit			

6 Operating cases

Case	Valve drop	Pump kW	Holds dest.
100% fouled	1.5 bar	13.12	Yes
Pump differential 8.08 bar · Kv 29.09			
100% clean	1.73 bar	13.12	Yes
Pump differential 8.08 bar · Kv 27.13			
120% fouled	0.62 bar	14.91	Yes
Pump differential 7.65 bar · Kv 54.3			
140% fouled	-0.42 bar	16.25	No
Pump differential 7.15 bar · Kv not sized			

60% clean	2.86 bar	8.48	Yes

Pump differential 8.7 bar · Kv 12.64

Notes

1. Pump, when sized, holds the destination pressure with fouled equipment and the control-valve drop entered on the valve.
2. Control-valve Cv is a liquid screen with piping factor 1. Body size places the design point near 70% travel on an equal-percentage valve with rangeability 50. Full-open Cv values are typical, not a catalog.
3. Clean and fouled cases keep the same pump differential. A centrifugal curve is a screen: design head at rated flow, about 12% higher at shutoff.
4. The sketch is a P&ID of the loop, not a scale plot plan. Pipe runs are the line between the symbols.
5. At 140% flow, fouled, the residual valve drop is gone. Destination pressure will not hold.
6. Screening study. Not issued for construction.

Prepared _____ Checked _____ Approved _____